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Nonbank Lending and Credit Cyclicalities

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1 Big picture

- **Question.** Who drives cyclical fluctuations in the supply of syndicated loans: banks or non-banks? Once nonbanks became large buyers of institutional term loans, can we still read swings in syndicated originations as primarily reflecting bank health?
- **Setting.** The paper studies the U.S. syndicated loan market from 2000Q1 to 2020Q4, with special attention to the Global Financial Crisis (GFC) and then to multiple credit cycles more broadly.
- **Key institutional fact.** Banks originate syndicated deals, but not all tranches are meant to stay on bank balance sheets. Revolvers and term loan As are mostly *bank* tranches, while term loan Bs (and term loans C–K) are *nonbank* tranches designed for institutional investors.
- **Main GFC result.** The paper argues that the 2008–2009 syndicated credit crunch was driven much more by the collapse in nonbank demand than by deterioration in bank health. In the aggregate decomposition, the reduction in nonbank deals accounts for about 71% of the lending decline, versus about 29% for bank deals.
- **Main real-outcome result.** Re-doing the Chodorow-Reich (2013) employment decomposition, the authors estimate that roughly 58% to 77% of credit-induced employment losses during the GFC are attributable to nonbank, not bank, credit availability.
- **Main multiple-cycle result.** Nonbank lending is much more cyclical than bank lending. In the within-deal specification, a one-standard-deviation increase in the Excess Bond Premium (EBP) reduces nonbank facility size by 13.9 percentage points more than bank facility size for the same borrower at the same time. The paper interprets this as nonbank credit supply being more than three times as cyclical as bank credit supply.
- **Price result.** When credit conditions tighten, nonbank spreads rise more than bank spreads. In the within-deal regressions, a one-standard-deviation increase in the EBP raises nonbank spreads by about 21 basis points more than bank spreads for the same deal.
- **Mechanism.** The paper’s baseline mechanism is that banks’ liabilities benefit from government support, while nonbank liabilities do not. When default risk rises, the relative funding cost of nonbanks rises and their lending contracts more sharply.
- **Why this matters.** The paper overturns a common reading of the syndicated-loan literature, which often treated bank health as the central driver of lending cycles. Once bank/nonbank segmentation is recognized, nonbanks emerge as the main marginal lenders in booms and busts.
- **Policy message.** A financial system with a larger nonbank sector may have less direct taxpayer support and less traditional bank risk, but it may also generate more cyclical aggregate credit supply.

2 Data and measurement (key objects)

2.1 Institutional setting and loan classification

- **Market studied.** The primary market for U.S. syndicated loans to nonfinancial firms. The paper argues this market is quantitatively large: by 2022, syndicated loans represented about

half of all loans outstanding to U.S. nonfinancial corporations and about 40% of the size of the nonfinancial corporate bond market.

- **Origination process.** Lead arrangers structure the deal, set spreads, and syndicate facilities to banks and nonbanks. Nonbanks often do not directly participate in the primary market for tax reasons and instead purchase institutional tranches immediately after origination.
- **Bank versus nonbank facilities.** Following industry practice and prior literature, the paper classifies term loan Bs through term loan Ks as *nonbank loans*, and all other facilities – especially credit lines and term loan As – as *bank loans*.
- **Deal structure.** A “deal” or “package” can include multiple facilities for the same borrower. This matters because some deals include both bank and nonbank tranches, letting the authors compare the two lender types for the same borrower at the same time.
- **Illustration.** Figure 1 shows a Yum! Brands deal with a \$2bn term loan B placed with nonbanks, alongside a \$0.5bn term loan A and a \$1bn revolver placed with banks. This is the institutional backbone of the identification strategy.

2.2 Data sources

- **DealScan.** Thomson Reuters DealScan provides facility-level data on newly originated and refinanced syndicated loans between 2000Q1 and 2020Q4. Refinancings are identified as amendments that change both spread and maturity; the paper treats these as new loans.
- **Compustat and CRSP.** Used to measure borrower opacity, ratings, leverage, and other time-varying borrower characteristics. Compustat is also used, via lender-bank matching files, to build bank capital ratios following Schwert (2018).
- **Bank health data for the GFC.** The paper uses the bank-health measures studied by Chodorow-Reich (2013), including Lehman exposure, ABX exposure, trading revenues, real estate indicators, and deposits-to-assets.
- **Credit-cycle measure.** The key aggregate shock is the Excess Bond Premium (EBP) from Gilchrist and Zakrajšek (2012), interpreted as the compensation investors require beyond expected default losses.
- **Nonbank holdings and composition.** Shared National Credit (SNC) data are used to describe the growing role of nonbanks, while SIFMA and Morningstar data help show the composition of nonbank lenders over time.
- **CLO data.** Creditflux data on CLO tranches and holdings are used to study CLO leverage; Palmer Square CLO debt indexes are used to measure CLO funding costs relative to bank funding costs.

2.3 Key variables from Table 1

- **Sample size.** The sample contains 79,198 bank facilities and 14,802 nonbank facilities.
- **Loan type composition.** Among bank loans, 84.46% are credit lines and other bank facilities, while 15.54% are term loan As. By construction, 100% of nonbank loans are term loan Bs (or institutional tranches).

- **Size and price.** Nonbank loans are larger on average (\$381m vs. \$277m) and more expensive (390 bps vs. 275 bps).
- **Maturity and risk.** Nonbank loans have longer maturity (66.7 vs. 48.7 months) and are much more likely to be leveraged loans (87% vs. 52%).
- **Overlap.** About 18% of bank facilities are part of deals that also contain a nonbank tranche. This overlap is crucial because it makes within-deal comparisons possible.
- **Loan purpose.** More than half of both bank and nonbank facilities finance general corporate purposes, so the comparison is not purely between unrelated products such as working-capital credit lines versus speculative buyout loans.

2.4 Core constructed measures

- **Nonbank dependence of a bank.** For bank b , NBDep_b is the fraction of loans it originates through *nonbank deals*, that is, deals containing at least one institutional tranche.
- **Bank deal versus nonbank deal.** A bank deal contains only bank-type facilities. A nonbank deal contains at least one nonbank tranche.
- **Real investment loans.** In some exercises the paper narrows to facilities whose purpose is “corporate purpose,” “working capital,” or “capital expenditure.”
- **Credit cycle.** The EBP is standardized to mean zero and standard deviation one. A higher EBP means tighter aggregate credit conditions.
- **Borrower opacity.** Small, young, private, and low-DealScan-history borrowers proxy for settings where information asymmetry may be stronger.

3 Models and empirical specifications (all equations + interpretation)

3.1 Decomposition of the GFC credit crunch

The paper first decomposes the change in loan originations by bank b into the contribution from bank deals and nonbank deals:

$$\Delta L_b = (1 - \text{NBDep}_b) \Delta L_b^B + \text{NBDep}_b \Delta L_b^{NB}. \quad (1)$$

Dividing by total lending change yields the share of the lending decline attributable to each segment:

$$1 = (1 - \text{NBDep}_b) \frac{\Delta L_b^B}{\Delta L_b} + \text{NBDep}_b \frac{\Delta L_b^{NB}}{\Delta L_b}. \quad (2)$$

- **Interpretation.** A bank’s total origination collapse can be split into the part coming from the loans it usually keeps on bank balance sheets and the part coming from the loans it usually distributes to nonbanks.
- **Why this matters.** If the collapse is concentrated in the second term, then a regression of total bank-level origination changes on bank-health measures will conflate bank weakness with nonbank withdrawal.

3.2 Bank health versus nonbank dependence during the GFC

The paper formalizes the omitted-variable problem as

$$\Delta L_b = \beta \cdot \text{BankHealth}_b + \text{NBDep}_b \cdot \psi^{NB}, \quad (3)$$

where ψ^{NB} is a common nonbank shock.

If one omits nonbank dependence, the bias in the bank-health coefficient is

$$\hat{\beta} - \beta = \psi^{NB} \cdot \frac{\text{Cov}(\text{BankHealth}_b, \text{NBDep}_b)}{\text{Var}(\text{BankHealth}_b)}. \quad (4)$$

- **Interpretation.** The standard bank-health regression is biased if weak banks are also the banks that rely most on nonbanks.
- **Sign intuition.** The paper finds exactly this pattern: weaker banks have greater nonbank dependence, and the nonbank shock in the GFC is very negative. That makes the naive bank-health coefficient too large in absolute value.
- **Economic message.** What earlier work read as “bank health caused the syndication collapse” is in part “banks exposed to nonbanks saw nonbanks disappear.”

3.3 Employment-loss decomposition

To revisit Chodorow-Reich (2013), the paper computes firm-level employment losses as

$$\Delta E_i = \zeta_{1,i} \cdot \min\{\Delta L_{i,s} - \Delta L_{Q_\tau}, 0\}, \quad (5)$$

where $\Delta L_{i,s}$ is firm i ’s exposure to the credit crunch through its precrisis lending syndicate, $\zeta_{1,i}$ maps lost credit into employment loss, and ΔL_{Q_τ} is the threshold below which a lending change is interpreted as a credit-supply shock.

- **Interpretation.** A firm loses employment only when its syndicate contracts more than a benchmark threshold.
- **Decomposition idea.** The paper then uses the bank/nonbank decomposition above to split $\Delta L_{i,s}$ into a bank component and a nonbank component.
- **Purpose.** This is how the authors move from lending effects to real effects.

3.4 Aggregate cyclicalities over multiple credit cycles

To compare bank and nonbank cyclicalities in the time series, the paper estimates

$$\text{LendingOutcome}_{ft} = \beta_0 + \beta_1 \text{CreditCycle}_{t-1} + \beta_2 \mathbf{1}_{f=\text{TermB}} + \beta_3 \text{CreditCycle}_{t-1} \times \mathbf{1}_{f=\text{TermB}} + \varepsilon_{ft}. \quad (6)$$

- **Dependent variables.** Either the log of aggregate issuance volume for facility type f in month t , or the average all-in-drawn spread for that type-month.
- **Interpretation of coefficients.** β_1 is the sensitivity of *bank* loan outcomes to the credit cycle; β_3 is the additional sensitivity of *nonbank* outcomes.

- **Expected signs.** For volumes, $\beta_1 < 0$ and $\beta_3 < 0$ if both contract in bad times and nonbanks contract more. For spreads, $\beta_1 > 0$ and $\beta_3 > 0$ if both reprice upward and nonbanks reprice more.
- **Limitation.** This is still partly reduced-form because bank and nonbank borrowers could differ in demand and risk.

3.5 Within-deal supply identification

The main identification equation is

$$\text{LendingOutcome}_{idft} = \delta_{idt} + \beta_1 \mathbf{1}_{f=\text{TermB}} + \beta_2 \text{CreditCycle}_{t-1} \times \mathbf{1}_{f=\text{TermB}} + \varepsilon_{idft}, \quad (7)$$

where δ_{idt} are borrower-by-deal-by-time fixed effects.

- **Interpretation.** This compares bank and nonbank facilities for the same borrower, in the same deal, at the same point in time.
- **Why it is powerful.** Since facilities in the same deal have the same seniority and are governed by the same contract, deal fixed effects absorb borrower demand and borrower risk that are common across tranches.
- **Coefficient of interest.** β_2 is the differential response of nonbank credit supply relative to bank credit supply when aggregate credit conditions tighten.
- **Economic meaning.** If nonbank facilities become smaller and more expensive than bank facilities within the same deal when EBP rises, the difference is hard to explain with borrower demand alone.

3.6 Within-bank and lender-role robustness specifications

To test whether the main result is really about nonbanks rather than weak arranging banks, the paper also estimates bank-level versions such as

$$\text{LendingOutcome}_{bft} = \delta_{bt} + \beta_1 \text{CreditCycle}_{t-1} \times \mathbf{1}_{f=\text{TermB}} + \beta_2 \mathbf{1}_{f=\text{TermB}} + \varepsilon_{bft}, \quad (8)$$

with either bank fixed effects or bank-by-month fixed effects.

It also replaces the bank/nonbank split with lender-role groups: lead banks, nonlead participant banks, and nonbanks.

- **Interpretation.** These designs ask whether nonbank tranches remain more cyclical even when they are originated by the same bank, or even when they are compared to nonlead participant banks that do not have special monitoring roles.
- **Why this matters.** If originating-bank health or lead-bank informational rents were the whole story, these tighter comparisons should wipe out the result.

3.7 Baseline model: banks

The paper's one-period model assumes banks and nonbanks use a Cobb–Douglas production technology with capital input Q_j , success probability $1 - \delta$, and failure probability δ .

Banks choose production scale Q_B to solve

$$\max_{Q_B} (1 - \delta) \left[Q_B^\alpha - (R_B + c)(Q_B - W_B) \right], \quad (9)$$

where W_B is bank net worth, R_B is the gross promised rate on bank liabilities, and c is the quasi-fixed per-unit cost of access to government support.

The first-order condition implies

$$Q_B = \left(\frac{\alpha}{R_B + c} \right)^{\frac{1}{1-\alpha}}. \quad (10)$$

- **Interpretation.** Banks lend more when their funding cost is lower. The cost of government support enters like an additional funding wedge.
- **Key point.** If government support makes R_B relatively stable over the cycle, bank lending becomes relatively stable too.

3.8 Baseline model: nonbanks

Nonbanks solve

$$\max_{Q_N} (1 - \delta) \left[Q_N^\alpha - R_N(Q_N - W_N) \right], \quad (11)$$

which yields

$$Q_N = \left(\frac{\alpha}{R_N} \right)^{\frac{1}{1-\alpha}}. \quad (12)$$

- **Difference from banks.** Nonbanks have no government guarantee. In the frictionless baseline, Modigliani–Miller holds, so the source of nonbank financing does not matter; what matters is that investors demand compensation for default risk.
- **Interpretation.** As nonbank funding costs rise, nonbank lending contracts.

3.9 Households and equilibrium funding costs

The representative household solves

$$u(C_0, C_1) = \max C_0 + \beta \mathbb{E}[C_1] \quad (13)$$

subject to

$$C_0 = W_H - D_B - D_N, \quad (14)$$

and

$$C_1 = \begin{cases} R_B D_B + R_N D_N, & \text{if production succeeds,} \\ R_B D_B, & \text{if production fails.} \end{cases} \quad (15)$$

The Euler equations are

$$R_B = \beta^{-1}, \quad (16)$$

and

$$R_N = \frac{\beta^{-1}}{1 - \delta}. \quad (17)$$

- **Interpretation.** Because bank debt is guaranteed, households require only the risk-free discount factor on bank liabilities. Because nonbank liabilities are not guaranteed, required returns rise with default probability δ .
- **Prediction 1.** Nonbank financing costs rise relative to bank financing costs when default probability increases.

3.10 Equilibrium lending and model predictions

Substituting equilibrium funding costs into the bank and nonbank first-order conditions gives

$$Q_B = \left(\frac{\alpha}{\beta^{-1} + c} \right)^{\frac{1}{1-\alpha}}, \quad (18)$$

and

$$Q_N = \left(\frac{\alpha(1-\delta)}{\beta^{-1}} \right)^{\frac{1}{1-\alpha}}. \quad (19)$$

- **Prediction 2.** Nonbank lending falls relative to bank lending as default risk rises, because only nonbank funding costs move one-for-one with δ .
- **Prediction 3.** For debt-financed nonbanks, leverage falls relative to bank leverage in bad times because quantities and debt issuance contract together.
- **Economic intuition.** In good times, the quasi-fixed cost c of government support can make banks relatively expensive. In bad times, that support becomes valuable and stabilizes their funding.

3.11 Financing-frictions extension

- **Additional amplification.** The appendix extends the model to allow run risk and balance-sheet constraints for nonbanks.
- **Interpretation.** Open-end loan funds can face run-type redemptions, while CLOs can face agency or balance-sheet frictions. If these frictions worsen in bad times, they make nonbank lending even more cyclical than in the frictionless benchmark.
- **Why this matters empirically.** The paper’s mechanism evidence on CLO funding costs and leverage is designed to speak to these model predictions.

4 Identification and instruments (what makes the estimates credible)

4.1 GFC identification: the omitted variable is nonbank dependence

- **Core identification idea.** The paper’s first move is not an instrument but a decomposition. It asks whether bank-health regressions are misspecified because they ignore variation in banks’ reliance on nonbanks.

- **Crucial fact.** Figure 3 shows that banks with weaker health before the crisis were also those most active in nonbank deals. This makes nonbank dependence correlated with the standard bank-health proxies.
- **Implication.** Once nonbank dependence is controlled for, the explanatory power of traditional bank-health proxies for *total* loan originations falls sharply and usually becomes insignificant.
- **What is identified.** The paper identifies that much of what looked like a bank-health channel in the GFC was really a joint bank/nonbank composition effect.

4.2 Multiple-cycle identification: same borrower, same deal, same time

- **Main design.** Compare bank and nonbank facilities issued to the same borrower, in the same deal, at the same time, under essentially the same seniority and contractual environment.
- **Why the deal fixed effect matters.** It absorbs borrower-level shocks to demand and default risk that are common across facilities in that deal.
- **Remaining identifying assumption.** Conditional on the fixed effects, there is no within-deal unobserved demand shock that selectively tilts borrowers toward term loan B facilities exactly when the EBP rises.
- **Why plausible.** Nonbank facilities generally fund *riskier* borrowers. If tightening conditions mainly increased borrower risk, one might expect more, not less, nonbank borrowing within the same deal. The observed pattern goes the opposite way.

4.3 No external IV in the main design

- **Important feature.** The central multiple-cycle result is not identified through an external instrument. It is identified through a matched-lender design that exploits segmentation inside syndicated deals.
- **Strength.** This avoids the need for a controversial instrument for bank or nonbank funding shocks.
- **Tradeoff.** Credibility depends on the institutional claim that bank and nonbank tranches within a deal face the same borrower fundamentals and demand conditions.

4.4 How the paper addresses alternative explanations

- **Originating-bank health.** Table 7 shows that within the same bank, nonbank originations are still much more cyclical than bank originations. So the effect is not just that weak banks originate more term loan Bs.
- **Lead-bank information and monitoring.** Table 8 compares nonbanks to *nonlead participant banks*, which do not enjoy the same special role in screening and monitoring. The nonbank cyclicity result remains.
- **Opacity heterogeneity.** Table 9 finds only limited support for the idea that the bank/nonbank gap is strongest precisely where information asymmetry should matter most.

- **Borrower composition.** The appendix adds ratings, volatility, coverage ratios, leverage, private-firm subsamples, real-investment-loan subsamples, alternative credit-cycle measures, term-loan-only samples, and samples excluding the GFC. The main result survives.

4.5 Mechanism identification

- **Theory-to-data mapping.** The model predicts three things: nonbank funding costs become more cyclical than bank funding costs, nonbank lending becomes more cyclical, and debt-financed nonbank leverage falls more in bad times.
- **Evidence used.** Figure 7 shows that CLO funding spreads over LIBOR rise when EBP rises, while bank equity ratios remain strikingly low even in bad times. This is consistent with banks benefiting from support that stabilizes their liabilities.
- **Cross-bank implication.** Banks with more deposit financing are slightly less likely to exit syndicates when EBP rises, consistent with stronger government support, though this cross-bank effect is small relative to the huge bank/nonbank gap.

5 Tables and figures

5.1 Figure 1: Example of a syndicated loan deal

Read:

- The figure gives the paper’s core institutional picture: one borrower, one deal, multiple facilities, and a split between bank-facing and nonbank-facing tranches.
- The key empirical point is that the bank/nonbank comparison is not across unrelated borrowers. It often happens *within the same deal*.
- This is why the paper can later use deal fixed effects to separate credit supply from credit demand.

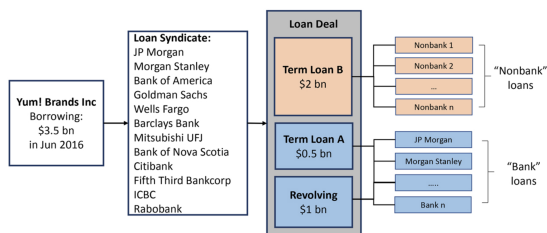


Figure 1
Example of a syndicated loan deal
This figure provides an example of a syndicated loan deal (LPC Deal ID = “209819”). JP Morgan functioned as “admin agent,” and all other syndicate banks functioned as “arrangers.”

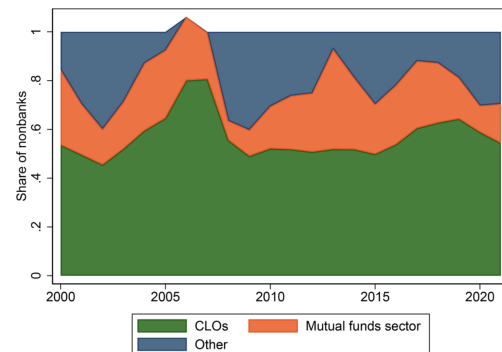


Figure 2
Composition of nonbank lenders
This figure shows the decomposition of nonbank lenders in the syndicated loan market. The assets under management of collateralized loan obligations (CLOs) are obtained from the Securities, Industry and Financial Markets Association (SIFMA). The assets under management of loan mutual funds (open-end and closed-end), ETFs, and separate accounts are obtained from Morningstar. The total nonbank holdings are from Shared National Credit (SNC) Reports (see <https://www.occ.treas.gov/publications-and-resources/publications/shared-national-credit-report/index-shared-national-credit-report.html>). Because of the reporting requirements, the SNC number represents a lower bound on the total nonbank loan holdings. This explains why the combined mutual fund and CLO holdings were higher than SNC-reported nonbank loan holdings in 2006. The sample period is from 2000 to 2020.

5.2 Figure 2: Composition of nonbank lenders

Read:

- CLOs are the dominant nonbank lenders, typically accounting for roughly one-half to four-fifths of nonbank holdings.
- Mutual funds, ETFs, and separate accounts form the next major block, often around one-fifth of the nonbank sector.
- The figure matters for the mechanism section because different nonbanks face different financing frictions: CLOs look like long-term debt-financed vehicles, while open-end funds can look more run-prone.

5.3 Table 1: Summary statistics

Read:

- Nonbank loans are larger, longer maturity, and substantially riskier than bank loans, and they carry spreads about 115 basis points higher on average.
- But the overlap is still meaningful: nearly one-fifth of bank facilities are in deals that also include a nonbank tranche.
- This table is important because it shows that bank and nonbank loans are different, but not so different that comparison is impossible.

Table 1
Summary statistics

A. Loan type

	Bank loans %	Nonbank loans %
Term loan A	15.54	0.00
Term loan B	0.00	100.00
Credit line & other loans	84.46	0.00
Observations	79,198	14,802

B. General loan characteristics

	Bank loans		Nonbank loans	
	Mean	SD	Mean	SD
Volume (in mill. USD)	277.29	807.30	380.99	624.45
Spread (in basis points)	274.74	165.63	389.62	139.80
Maturity (in months)	48.74	22.59	66.65	18.27
Leveraged loan (in %)	52.07	49.96	87.02	33.61
Part nonbank deal (in %)	17.98	38.41	100.00	0.00
Observations	79,198		14,802	

C. Loan purpose

	Bank loans %	Nonbank loans %
General purpose	58.85	54.17
Working capital	8.00	2.05
LBO	3.31	19.10
Other	29.84	24.68
Observations	79,198	14,802

This table presents the summary statistics at the loan level for the loan-volume-weighted loan types (panel A), general loan characteristics (panel B), and the loan-volume-weighted loan purposes (panel C). *Nonbank loans* are all loan facilities classified as Term Loan B-Term Loan K. *Bank loans* are all other loan types. *Volume* is the loan facility size. *Spread* is the all-in-drawn-spread of the facility. *Maturity* is the facility-level maturity. *Leveraged loan* indicates whether the spread is above 225, following the leveraged loan definition of Lee et al. (2019). *Part nonbank deal* indicates whether the loan facility is part of a loan deal that contains a nonbank loan facility. Loans are classified as *General purpose* if the loan purpose variable in DealScan contains "General Purpose," *Working capital* if the loan purpose contains "Working Capital," and *LBO* if the loan purpose contains "Buyout." The sample period is from January 2000 and December 2020.

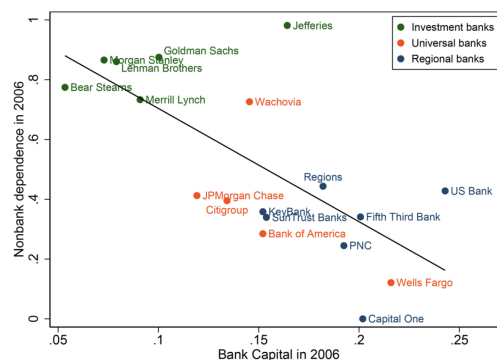


Figure 3
Bank-nonbank matching pattern

This figure shows the sorting pattern between banks and nonbanks. It shows the relationship between a lead arranger's nonbank dependence in 2006 and (a) its business model (investment bank, universal bank or regional bank), and (b) its capital ratio in 2006. Nonbank dependence is defined as the fraction of loans originated as part of nonbank deals. A nonbank deal is a loan deal that contains a nonbank tranche, that is, Term Loan B-K. Each loan (deal) is weighted by the share of the lender in the syndicate following Chodorow-Reich (2013). The aggregation of DealScan lender entities to bank holding companies follows Schwert (2018). Capital ratios are based on market equity relative to book liabilities and market equity obtained from Compustat following Schwert (2018). Only U.S. banks are included. More information on the sample construction can be found in the Section A1.2.1 of the Internet Appendix.

5.4 Figure 3: Bank-nonbank matching pattern

Read:

- The figure shows a clear negative relationship between bank capital and nonbank dependence in 2006.
- Investment banks are the most exposed to nonbank deals; regional banks are the least exposed.
- This figure is the central visual for the paper’s reinterpretation of the GFC bank-health literature: weaker banks were exactly the ones whose origination business depended more on nonbanks.

5.5 Table 2: A decomposition of the credit crunch during the GFC

Read:

- Across all banks, nonbank deals collapse by about 96% during the GFC, while bank deals fall by about 40%.
- Because precrisis nonbank dependence is about 50% in the aggregate, the nonbank side explains about 71% of the overall origination collapse.
- Read this table as the paper’s accounting core: the market-wide crunch was mostly a nonbank crunch.

Table 2

A decomposition of the credit crunch during the GFC

	Δ Loans in all deals	Δ Loans in nonbank deals	Nonbank depend.	Δ Loans in bank deals	Bank depend.	Nonbank contribution	Bank contribution	Market share
(1)=(2)·(3)+(4)·(5)	(2)	(3)	(4)	(5)	$= \frac{(2) \cdot (3)}{(1)}$	$= \frac{(4) \cdot (5)}{(1)}$		
Bank of America	-0.60	-0.94	0.36	-0.41	0.64	0.56	0.44	0.132
JPMorgan	-0.76	-0.98	0.42	-0.59	0.58	0.55	0.45	0.102
GE Capital	-0.82	-0.97	0.75	-0.39	0.25	0.88	0.12	0.075
Wachovia	-0.75	-0.99	0.43	-0.57	0.57	0.57	0.43	0.056
Wells Fargo	-0.32	-0.87	0.40	0.05	0.60	1.09	-0.09	0.052
Credit Suisse	-0.88	-0.95	0.83	-0.55	0.17	0.89	0.11	0.047
Citigroup	-0.74	-0.89	0.49	-0.59	0.51	0.59	0.41	0.046
CIT	-0.80	-0.94	0.78	-0.29	0.22	0.92	0.08	0.034
PNC	-0.31	-0.96	0.26	-0.08	0.74	0.81	0.19	0.028
Goldman Sachs	-0.92	-0.98	0.79	-0.69	0.21	0.84	0.16	0.027
All banks	-0.68	-0.96	0.50	-0.40	0.50	0.71	0.29	1.000

This table uses Equations (1) and (2) to decompose the change in the number of loans originated by a bank during the financial crisis. The lending change during the GFC is the change in the number of loans originated by a bank between the period October 2005 to June 2007 and October 2008 to June 2009. Nonbank dependence is defined as the fraction of loans that are originated as part of nonbank deals. A nonbank deal is a loan deal that contains a nonbank loan as defined in Section 2. Nonbank dependence has been computed for a period before the financial crisis, from October 2005 to June 2007. Each loan is weighted by the share of the lender in the syndicate following Chodorow-Reich (2013). The aggregation of DealScan lender entities to bank holding companies follows Chodorow-Reich (2013). The table shows the lending changes of the top 10 banks by precrisis market share, as well as the lending change of all banks (inside and outside of the top 10).

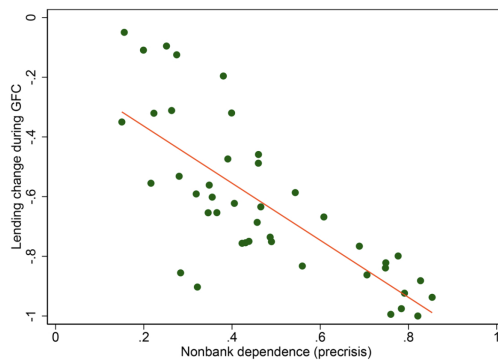


Figure 4

Bank-nonbank matching and lending in the GFC

This figure documents the relationship between precrisis nonbank dependence and the decline in loan origination volume during the GFC. Nonbank dependence is defined as the fraction of loan volume that is originated as part of nonbank deals. A nonbank deal is a loan deal that contains a nonbank tranche, that is, Term Loan B-K. Nonbank dependence is computed prior to the financial crisis for the period from October 2005 to June 2007. The lending change during the GFC is the volume change of loans originated by a bank between the period October 2005 to June 2007 and October 2008 to June 2009. The aggregation of DealScan lender entities to bank holding companies follows Chodorow-Reich (2013). Each loan is weighted by the estimated share of the lender in the syndicate following Chodorow-Reich (2013).

5.6 Figure 4: Bank-nonbank matching and lending in the GFC

Read:

- The scatterplot shows a strong negative relationship between precrisis nonbank dependence and GFC lending growth.
- Banks more exposed to nonbank deals before the crisis experienced much larger origination collapses during the crisis.
- This visual links Figures 3 and Table 2: the sorting pattern in normal times becomes the mechanism through which nonbanks magnify the credit crunch in bad times.

5.7 Table 3: Bank health versus nonbank dependence during the GFC

Read:

- Replicating Chodorow-Reich-type regressions without nonbank dependence gives significant bank-health effects.
- Once nonbank dependence is added, the R^2 rises materially and the bank-health coefficients lose significance, while the coefficient on nonbank dependence is around -0.83 to -0.96 .
- The bank-health proxies still help explain *bank deals*, but not *nonbank deals*. That is exactly the decomposition the paper wants the reader to internalize.

Table 3
Bank health versus nonbank dependence during the GFC

	Δ All loans						Δ Bank	Δ Nonbank
	(1)	(2)		(4)		(6)	(7)	(8)
Lehman exposure	-0.136*** (0.034)			0.029 (0.022)			-0.098 (0.077)	0.006 (0.024)
ABX exposure		-0.097* (0.048)			-0.036 (0.046)		-0.061 (0.067)	-0.003 (0.013)
07-08 trading rev/AT			0.061** (0.029)			0.031 (0.027)	0.114*** (0.041)	-0.007 (0.014)
RE CO flag			0.011 (0.049)			-0.057 (0.046)	-0.061 (0.055)	-0.010 (0.015)
07-08 RE NCO/AT			-0.081* (0.043)			-0.036 (0.039)	-0.061 (0.050)	-0.001 (0.012)
07 deposits/assets			0.175*** (0.054)			0.068 (0.059)	0.101 (0.083)	0.015 (0.024)
Nonbank dependence				-0.964*** (0.176)	-0.830*** (0.150)	-0.901*** (0.243)		
Obs.	42	40	42	42	40	42	38	38
R ²	0.171	0.136	0.344	0.458	0.490	0.520	0.274	0.033

This table reports regression results on lending changes during the GFC. The unit of observation is a bank. Columns 1–6 considers all loans, while columns 7 and 8 include only bank and nonbank loan deals, respectively. The lending change is the change in the number loans originated by a bank between the period October 2005 to June 2007 and October 2008 to June 2009. Nonbank dependence is defined as the fraction of loans that are originated as part of nonbank deals. A nonbank deal is a loan deal that contains a nonbank loan (Term Loan B-K) as defined in Section 2. Nonbank dependence has been computed for a period before the financial crisis, from October 2005 to June 2007. Bank health measures have been obtained from Chodorow-Reich (2013) and have been normalized to have mean zero and standard deviation of one. Each loan is weighted by the share of the lender in the loan syndicate computed as in Chodorow-Reich (2013). The observations are weighted by the number of precrisis borrowers to capture the economic importance of each bank. Robust standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4
Employment effects of bank and nonbank credit availability

	Credit supply percentile		
	$\tau = 90$ (1)	$\tau = 95$ (2)	$\tau = 98$ (3)
Total employment decline 2008:3 – 2009:3 (%)	7.0	7.0	7.0
Credit supply threshold L_{Q_τ} (%)	-29.7	-19.5	0.0
Share of losses due to credit availability (%)	36.6	49.5	77.1
thereof: Nonbank credit availability (%)	77.4	69.5	58.2
thereof: Bank credit availability (%)	22.6	30.5	41.8

This table reports the fraction of employment losses due to *nonbank* and *bank* credit availability at small and medium firms. We follow the methodology and estimates in Chodorow-Reich (2013). Firm i 's employment loss is computed as

$$\Delta E_i = \zeta_{1,i} \cdot \min(\Delta L_{i,s} - \Delta L_{Q_\tau}, 0),$$

where $\Delta L_{i,s}$ measures firm i 's exposure to the credit crunch by looking at how much its precrisis syndicate members (in the last loan obtained by the firm i prior to the GFC) reduced their lending for all other borrowers. $\zeta_{1,i}$ measures the pass-through from credit to employment that depends on the size of firm i (smaller firms are more affected than medium-sized firms, large firms are not affected). τ refers to the percentile of the lending syndicate that is assumed to not have changed its credit supply or "lending function." Accordingly, ΔL_{Q_τ} states the threshold below which a lending change is ascribed to a decline in credit supply. We then use Equation (1) to decompose $\Delta L_{i,s}$ into a *nonbank* component and a *bank* component. Summing across all small and medium firms and using the estimates for $\zeta_{1,i}$ from Chodorow-Reich (2013), we obtain the aggregate employment losses due to *nonbank* and *bank* credit availability. The methodology is described in detail in Section A1.2.4 of the Internet Appendix.

5.8 Table 4: Employment effects of bank and nonbank credit availability

Read:

- Depending on how much of the overall contraction is attributed to credit supply, between 36.6% and 77.1% of aggregate employment losses are linked to credit availability.
- Of those credit-induced losses, about 58.2% to 77.4% are attributed to nonbank credit availability, versus 22.6% to 41.8% for bank credit availability.
- This table takes the paper beyond the syndicated-loan market itself: the nonbank channel matters for real employment outcomes.

5.9 Figure 5: Cyclicalities of bank versus nonbank originations

Read:

- Bank originations are relatively smooth over time, while nonbank originations are visibly boom-bust.
- The GFC collapse is much sharper for nonbanks, but the same qualitative pattern appears outside the crisis too.

- The contrast is even stronger for loans tied to real investment, which strengthens the interpretation that this is not just about refinancing or buyout activity.

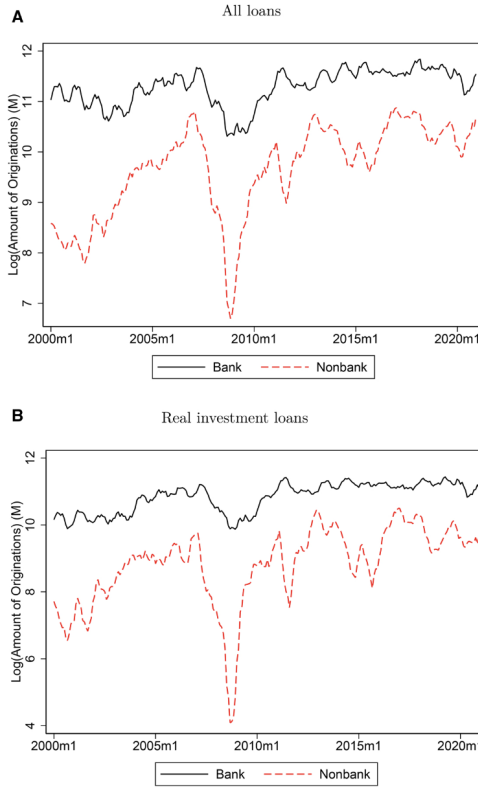


Figure 5
Cyclicality of bank versus nonbank originations
This figure shows new originations of bank and nonbank loans between January 2000 and December 2020. We plot a 6-month (forward-looking) average of the logarithm of the total origination amount for bank and nonbank loans. Nonbank loans are loans classified as Term Loan B-K, while bank loans are all other loans, including credit lines, Term Loan As or undefined term loans, in DealScan. Panel A contains all loans, while panel B includes only real investment loans. Real investment loans are loans whose primary purpose is defined as “corporate purpose,” “working capital,” or “capital expenditure” according to DealScan.

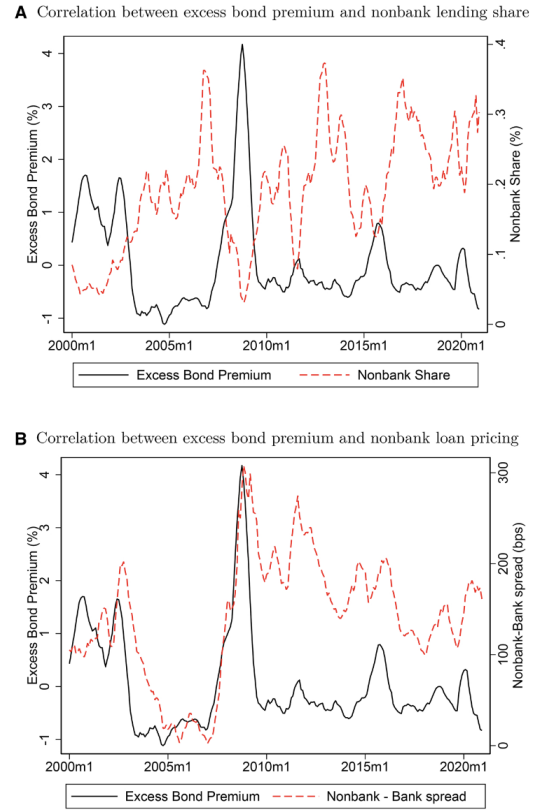


Figure 6
Relative sensitivity of bank and nonbank lending to the credit cycle
This figure shows nonbank lending varies with aggregate credit conditions, relative to bank lending. Panel A shows the nonbank share of newly originated loans versus the Excess Bond Premium from Gilchrist and Zakrajšek (2012) as a measure of the credit cycle. Panel B shows the difference in the all-in-drawn spread between nonbank and bank term loans, after controlling for loan maturity, seniority, loan purpose and borrower industry. All series are smoothed using a 6-month (forward-looking) average. Nonbank loans are loans classified as Term Loan B-K, while bank loans are all other loans, including credit lines, Term Loan As or undefined term loans, in DealScan. The sample period is from January 2000 to December 2020.

5.10 Figure 6: Relative sensitivity of bank and nonbank lending to the credit cycle

Read:

- Panel A shows a strong inverse comovement between the EBP and the nonbank lending share.
- Panel B shows that the nonbank-minus-bank spread widens when the EBP rises.
- The figure is valuable because it shows the paper’s main result is not just a GFC story; the relationship is visible repeatedly over small and large credit-cycle fluctuations.

5.11 Table 5: Nonbank lending cyclicalities – aggregate results

Read:

- In the aggregate, a one-standard-deviation increase in EBP is associated with a 44.0% decline in total originations.
- Splitting lender types, the implied decline is about 16.9% for bank loans versus about 71.2% for nonbank loans; spreads rise about 12.7 bps for banks versus 31.7 bps for nonbanks.
- This is the first formal evidence that nonbank lending is much more cyclical, but it still leaves open demand-composition concerns.

Table 5
Nonbank lending cyclicity: Aggregate results

A. Loan volumes			
	log(Facility amount)		
	(1)	(2)	(3)
Excess bond premium	-0.440*** (0.042)	-0.169*** (0.037)	
Term B	-1.653*** (0.059)	-1.629*** (0.054)	-1.629*** (0.038)
Excess bond premium x Term B		-0.543*** (0.062)	-0.543*** (0.047)
Year-Month FE	N	N	N
Obs.	504	504	504
R ²	0.665	0.721	0.930
B. Loan spreads			
	All-in-drawn spread		
	(1)	(2)	(3)
Excess bond premium	22.208*** (4.449)	12.672*** (4.480)	
Term B	185.024*** (5.728)	184.182*** (5.569)	184.182*** (3.885)
Excess bond premium x Term B		19.072** (8.969)	19.072** (7.864)
Year-month FE	N	N	N
Obs.	504	504	504
R ²	0.688	0.695	0.923

This table reports results on aggregate monthly loan originations. The unit of observation is a loan-type x month pair. We report the results of

$$\text{Lending Outcome}_{f,t} = \beta_0 + \beta_1 \text{Credit Cycle}_{t-1} + \beta_2 \mathbb{1}_{f=\text{TermB}} + \beta_3 \text{Credit Cycle}_{t-1} \times \mathbb{1}_{f=\text{TermB}} + \epsilon_{f,t},$$

where Lending Outcome_{f,t} is either the logarithm of the total origination amount (panel A) or the weighted average spread (panel B) for all loans of facility type *f* originated in month *t*. $\mathbb{1}_{f=\text{TermB}}$ is a dummy variable that takes a value of 1 if the facility is a nonbank loan, as defined in Section 2. Credit cycle is the Excess Bond Premium from Gilchrist and Zakrajsek (2012). Our sample includes all loans originated between January 2000 and December 2020. The Excess Bond Premium is standardized to have mean 0 and standard deviation of 1. Robust standard errors are presented. * *p* < .1; ** *p* < .05; *** *p* < .01.

Table 6
Nonbank lending cyclicity: Within-deal results

A. Loan volumes					
	log(Facility amount)				
	(1)	(2)	(3)	(4)	(5)
Excess bond premium	-0.068*** (0.015)	-0.055*** (0.013)			
Term B	0.479*** (0.027)	0.445*** (0.029)	0.412*** (0.028)	0.585*** (0.038)	
Excess bond premium x Term B		-0.125*** (0.032)	-0.153*** (0.029)	-0.235*** (0.050)	-0.139*** (0.029)
Borrower FE	Y	Y	Y	N	N
Year-month FE	N	N	Y	N	N
Deal FE	N	N	N	Y	Y
Borrower x Facility-type FE	N	N	N	N	Y
Obs.	59,881	59,881	59,881	18,166	11,978
R ²	0.722	0.723	0.758	0.815	0.930
B. Loan spreads					
	All-in-drawn spread				
	(1)	(2)	(3)	(4)	(5)
Excess bond premium	16.041*** (2.380)	12.332*** (2.492)			
Term B	27.161*** (3.057)	36.363*** (2.665)	37.966*** (2.637)	0.170 (2.782)	
Excess bond premium x Term B		32.979*** (3.292)	30.547*** (2.652)	25.240*** (2.967)	21.114*** (2.811)
Borrower FE	Y	Y	Y	N	N
Year-month FE	N	N	Y	N	N
Deal FE	N	N	N	Y	Y
Borrower x Facility-type FE	N	N	N	N	Y
Obs.	53,452	53,452	53,452	16,576	10,882
R ²	0.678	0.680	0.719	0.786	0.904

This table reports results at the loan-type × loan deal level. We estimate

$$\text{Lending outcome}_{i,d,f,t} = \delta_{i,d,t} + \beta_1 \text{Credit cycle}_{t-1} \times \mathbb{1}_{f=\text{TermB}} + \beta_2 \mathbb{1}_{f=\text{TermB}} + \epsilon_{i,d,f,t},$$

where Lending outcome_{i,d,f,t} is the logarithm of the loan issuance volume (panel A), or the all-in-drawn spread (panel B) at origination to borrower *i* for deal *d*, facility-type *f*, which is either a bank loan or nonbank loan as defined in Section 2, in month *t*. $\mathbb{1}_{f=\text{TermB}}$ is a dummy variable that takes a value of 1 if the facility is a nonbank loan, as defined in Section 2. Credit cycle is the Excess Bond Premium from Gilchrist and Zakrajsek (2012). Our sample includes all loans originated between January 2000 and December 2020. The Excess Bond Premium is standardized to have mean 0 and standard deviation of 1. Standard errors are double clustered at the firm and month levels. * *p* < .1; ** *p* < .05; *** *p* < .01.

5.12 Table 6: Nonbank lending cyclicity – within-deal results

Read:

- This is the paper's main causal table. In the tightest quantity specification, a one-standard-deviation increase in EBP reduces nonbank facility size by 13.9 percentage points more than bank facility size for the same borrower in the same deal.
- In the corresponding spread specification, nonbank spreads rise by about 21.1 basis points more than bank spreads.
- The within-deal comparison is what lets the authors interpret the bank/nonbank gap as a differential credit-supply effect rather than a demand or borrower-risk effect.

5.13 Table 7: Within-bank cyclicity

Read:

- Even within the same originating bank, nonbank tranches are much more cyclical than bank tranches.

- In the bank-by-month fixed effect specification, the interaction is around -0.387 for quantities and about $+18.3$ bps for spreads.
- This directly rejects the idea that the paper is merely picking up cyclicalities of weak originating banks.

Table 7
Within-bank cyclicalities
A. Loan volume

	log(Amount)			
	(1)	(2)	(3)	(4)
Excess bond premium	-0.205*** (0.034)	-0.208*** (0.035)		
Term B	-1.477*** (0.140)	-1.728*** (0.119)	-1.815*** (0.123)	-1.878*** (0.112)
Excess bond premium x Term B	-0.278*** (0.059)	-0.337*** (0.055)	-0.387*** (0.048)	-0.372*** (0.055)
Bank FE	N	Y	N	N
Bank x Month FE	N	Y	N	N
Role	All	All	All	Nonlead
Obs.	17,086	17,086	14,442	13,826
R ²	0.227	0.573	0.844	0.834

B. Loan spread

	All-in-drawn-spread			
	(1)	(2)	(3)	(4)
Excess bond premium	11.785*** (3.581)	11.978*** (3.576)		
Term B	169.231*** (6.450)	167.485*** (6.415)	169.908*** (6.271)	169.139*** (6.042)
Excess bond premium x Term B	15.964*** (5.066)	15.534*** (4.891)	18.320*** (4.755)	18.265*** (5.027)
Bank FE	N	N	Y	N
Bank x Month FE	N	N	Y	N
Role	All	All	All	Non-Lead
Obs.	17,086	17,086	14,442	13,826
R ²	0.450	0.508	0.811	0.819

This table presents results on bank versus nonbank loan originations at the bank level. Observations are at the bank x loan-type x month level. We report results from the regression

$$\text{Lending outcome}_{bft} = \delta_{bt} + \beta_1 \text{Credit cycle}_{t-1} \times \mathbb{I}_{f=\text{TermB}} + \beta_2 \mathbb{I}_{f=\text{TermB}} + \epsilon_{bft},$$

where $\text{Lending outcome}_{bft}$ is either the logarithm of total loan amount (panel A) or the weighted average spread (panel B) of all loans of facility type f originated by bank b in month t . We distribute loan amount across all syndicate members within a facility following Chodorow-Reich (2013). $\mathbb{I}_{f=\text{TermB}}$ is a dummy variable that takes a value of 1 if the facility is a nonbank loan, as defined in Section 2. Credit cycle is the Excess Bond Premium from Gilchrist and Zakrajšek (2012). Our sample includes all loans originated between January 2000 and December 2020. Columns 1–3 include all loans. Column 4 considers only loans where bank b was not a lead arranger. The Excess Bond Premium is standardized to have mean 0 and standard deviation of 1. Standard errors are clustered at the bank and month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Cyclicalities of lead banks, nonlead banks, and nonbanks

	log(Facility amount)		
	(1)	(2)	(3)
Excess bond premium	-0.042*** (0.013)		
Lead bank x Excess bond premium	0.012 (0.008)	0.013 (0.008)	0.007 (0.008)
Nonbank x Excess bond premium	-0.141*** (0.035)	-0.164*** (0.032)	-0.229*** (0.048)
Borrower FE	Y	Y	N
Year-month FE	N	Y	N
Deal FE	N	N	Y
Lender-role FE	Y	Y	Y
Obs.	114,771	114,771	107,006
R ²	0.688	0.715	0.854

This table presents results on new loan originations as in Table 6 but by lender role. We distribute loan amount across all syndicate members within a facility following Chodorow-Reich (2013). Lead Banks denotes the amount of the bank loan held by lead arrangers of the syndicate. Nonlead bank amounts are based on the amount of bank loans held by nonlead banks. Nonbank amount is nonbank loan volume. Bank and nonbank loans are classified as described in Section 2. Credit cycle is the Excess Bond Premium from Gilchrist and Zakrajšek (2012), standardized to mean 0 and standard deviation 1. The lead bank is defined at the deal level and as in Sufi (2007) and Ivashina (2009). Nonlead banks are the omitted group. Our sample includes all loans originated between January 2000 and December 2020. Standard errors are double clustered at the firm and month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.14 Table 8: Cyclicalities of lead banks, nonlead banks, and nonbanks

Read:

- The nonbank result survives when the benchmark is *nonlead participant banks*, not just lead banks.
- In the tightest specification, nonbanks reduce lending by about 22.9 percentage points more than nonlead banks when EBP rises by one standard deviation.
- This weakens the alternative story that the results are driven by special lead-bank monitoring or “skin in the game” requirements.

5.15 Table 9: Cyclicalities by borrower characteristics

Read:

- The bank/nonbank cyclicalities gap is somewhat larger for small firms, but it does not become systematically larger for young or more opaque firms.
- For private firms, the relative nonbank cyclicalities is if anything smaller.
- Read this table as a partial defeat for the information-asymmetry explanation: opacity does not line up cleanly with the strength of the main result.

Table 9
Cyclicality of lead banks, nonlead banks, and nonbanks by borrower characteristics
Log(Facility Amount)

	(1)	(2)	(3)	(4)
Lead Bank $\times$ EBP	0.063*** (0.013)	0.038*** (0.012)	0.023** (0.010)	-0.015** (0.008)
Nonbank $\times$ EBP	-0.177*** (0.053)	-0.244*** (0.061)	-0.200*** (0.046)	-0.170*** (0.060)
Lead Bank $\times$ Small $\times$ EBP	-0.024* (0.014)			
Nonbank $\times$ Small $\times$ EBP	-0.150*** (0.054)			
Lead Bank $\times$ Young $\times$ EBP		0.025* (0.014)		
Nonbank $\times$ Young $\times$ EBP		-0.025 (0.066)		
Lead Bank $\times$ Young-DS $\times$ EBP			-0.040*** (0.013)	
Nonbank $\times$ Young-DS $\times$ EBP			-0.059 (0.046)	
Lead Bank $\times$ Public $\times$ EBP				0.065*** (0.012)
Nonbank $\times$ Public $\times$ EBP				-0.094* (0.051)
Borrower FE	N	N	N	N
Year-month FE	N	N	N	N
Deal FE	Y	Y	Y	Y
Lender-role FE	Y	Y	Y	Y
Obs.	44,372	45,035	107,006	91,109
R ²	0.873	0.857	0.865	0.866

This table presents results on new loan originations as in Table 6 but by lender role and borrower characteristics. We distribute loan amount across all syndicate members within a facility following Chodorow-Reich (2013). Lead Banks denotes the amount of the *bank loan* held by lead arrangers of the syndicate. Nonlead bank amounts are based on the amount of *bank loans* held by nonlead banks. Nonbank amount is nonbank loan volume. Bank and nonbank loans are classified as described in Section 2. Credit cycle is the Excess Bond Premium from Gilchrist and Zakrajšek (2012), standardized to mean 0 and standard deviation 1. The lead bank is defined at the deal level and as in Sufi (2007) and Ivashina (2009). Nonlead participant banks are the omitted group. *Small* indicates whether the borrower's total assets are below the median among all matched DealScan-Compustat firms. *Young* indicates whether the borrower's time since the first filing in Compustat is below the median. *Young-DS* indicates whether the borrower's number of loan deals in DealScan is below the median. *Public* indicates whether the borrower is in Compustat. Our sample includes all loans originated after 2000 in the matched DealScan-Compustat sample, which ends in 2017Q1. Standard errors are double clustered at the firm and month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

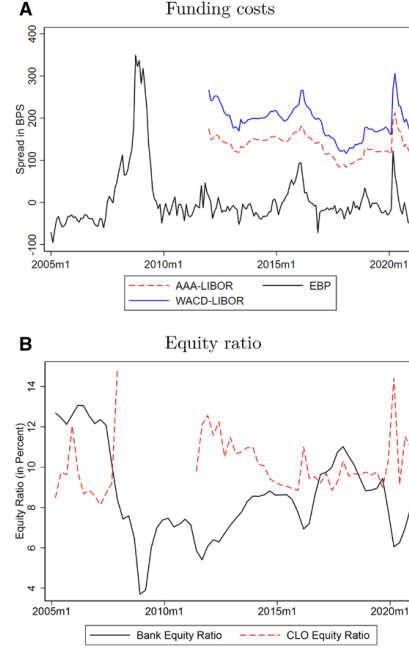


Figure 7
Banks versus CLOs: Funding costs and equity ratios
This figure plots the difference between CLO and bank funding costs (panel A) as well as the average CLO and bank equity ratios (panel B). **Funding costs:** The spread between CLO and bank funding costs is defined as the discount margin of outstanding CLO debt relative to the 3-month USD LIBOR rate. CLO discount margins are obtained from Palmer Squares CLO Debt Indexes, available through Bloomberg. The weighted average cost of debt (WACD) weights each CLO debt tranche-level index by the tranche's share in the average CLO capital structure. **Equity ratios:** CLO equity ratio is defined as the weighted average equity ratio of all newly issued CLOs in a month. CLO equity ratio is the size of the equity tranche relative to the combined notional value of CLO tranches at origination. We drop CLO equity ratio observations below and above the 2.5th and 97.5th percentiles, respectively, which are likely data errors. We also require at least two CLOs for a month to be included to avoid results driven by outliers. There were no CLO issuances in the Creditflux data from August 2008 to March 2010. We smooth the series by using the moving average of the current and next month. Bank equity ratio is defined as the weighted average $\frac{\text{book value of assets} + \text{market value of equity} - \text{book value of equity}}{\text{market value of equity}}$ of all bank holding companies in the DealScan-Compustat link file from Schwert (2018) in a given quarter. We take the moving average of the current and next quarter, and then interpolate between quarters to obtain a monthly series. For the year 2020, in which no DealScan-Compustat lender link file is available, we assume that the links from the prior year are still valid. Investment banks as classified in DealScan are excluded, and each bank's equity ratio is weighted by its lending market share in DealScan in the prior quarter. The sample period is January 2012 (when Palmer Square CLO debt indices become available) to December 2020 in panel A, and January 2005 (when Creditflux starts covering a large part of the CLO market) to December 2020 in panel B.

5.16 Figure 7: Banks versus CLOs – funding costs and equity ratios

Read:

- CLO funding spreads over LIBOR rise when EBP rises, even for very safe CLO debt tranches.
- At the same time, bank equity ratios remain strikingly low in bad times, while CLO equity ratios rise relative to bank equity. That is consistent with bank leverage being stabilized by government support.
- This figure ties the theory to the data: the more cyclical lender type also has more cyclical funding costs and balance-sheet leverage.

6 Short summary

- **Conclusion.** The paper shows that nonbanks, not banks, are the main drivers of cyclical fluctuations in syndicated lending.

- **Intuition.** Banks originate many loans for distribution to nonbanks. When nonbank demand disappears, bank-level origination collapses even if the bank-health channel is modest.
- **Empirical lesson.** In segmented credit markets, lender composition matters as much as lender health. Ignoring who ultimately funds the loan can lead to badly misleading causal interpretations.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The syndicated lending cycle is predominantly a *nonbank* lending cycle.
- **GFC lesson.** The paper reinterprets the 2008–2009 credit crunch: the collapse of nonbank deals, not simply weak bank balance sheets, explains most of the reduction in originations and much of the employment fallout.
- **Multiple-cycle lesson.** Across 2000–2020, nonbank credit supply is far more cyclical than bank credit supply, both in quantities and in prices.
- **Mechanism.** Government support stabilizes bank liabilities. Nonbanks lack that support and often face financing frictions that become more severe in bad times.
- **Broader implication.** A larger nonbank financial sector can amplify aggregate credit fluctuations even if it reduces the direct footprint of insured banks.

7.2 Economic intuition

- Banks and nonbanks coexist because they have different funding technologies. In normal times, nonbanks can be competitive or even cheap providers of credit, especially for institutional term loans.
- In bad times, however, the lack of guarantees makes nonbank funding more expensive precisely when default risk rises. Banks, by contrast, benefit from deposit insurance, emergency support, and other stabilizing backstops.
- Once one recognizes that banks often originate loans on behalf of nonbanks, the usual bank-health narrative changes: the marginal lender in many syndicated deals is not the bank arranging the loan but the nonbank expected to absorb the institutional tranche.

7.3 Takeaway

This paper’s core contribution is conceptual as much as empirical. It teaches that in modern credit markets, the identity of the *ultimate* lender matters for how credit responds to shocks. By separating bank tranches from nonbank tranches, Fleckenstein, Gopal, Gutiérrez, and Hillenbrand show that nonbanks are the main cyclical margin in syndicated lending and that government support is a plausible reason banks look relatively stable. The result is a major reinterpretation of both the GFC syndicated-loan evidence and the broader credit-cycle literature.